

## LUXOR INTL, Egypt

WMO: 624050

Lat: 25.671N

Lon: 32.707E

Elev: 90

StdP: 100.25

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | 6.0        | 7.2 | -2.7                          | 3.1 | 16.7 | -1.1 | 3.5 | 15.0 | 7.4                   | 18.0 | 6.5 | 18.2 | 0.9                   | 160  | 0.392 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 14.6                   | 43.8            | 23.4 | 42.5 | 23.0 | 41.4 | 22.7 | 24.8                | 40.4 | 24.2 | 39.8 | 23.6 | 39.2 | 2.9                  | 330  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 19.8                            | 14.7 | 33.9 | 18.8 | 13.8 | 34.0 | 17.9 | 13.0 | 34.1 | 75.8          | 40.4 | 73.0 | 39.9 | 70.9 | 39.3 | 28.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |      |            |      |            |      |      |  |
|-------------------|------|-----|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|------|------------|------|------------|------|------|--|
|                   |      |     | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |      | n=20 years |      | n=50 years |      |      |  |
| 1%                | 2.5% | 5%  | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max  | Min        | Max  | Min        | Max  |      |  |
| (a)               | (b)  | (c) | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |  |
| 6.8               | 5.9  | 5.1 | DB                         | 3.2 | 46.3               | 1.4 | 1.5                                                | 2.2 | 47.4       | 1.4  | 48.3       | 0.7  | 49.1       | -0.3 | 50.2 |  |
|                   |      |     | WB                         | 1.2 | 26.3               | 1.0 | 1.1                                                | 0.5 | 27.1       | -0.1 | 27.7       | -0.7 | 28.4       | -1.5 | 29.1 |  |

## Monthly Climatic Design Conditions

|                                                                   |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                   |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 25.7   | 15.1  | 17.3  | 21.4  | 26.4  | 30.7  | 33.0  | 33.7  | 33.5  | 31.4  | 27.5  | 21.3  | 16.5  |
|                                                                   | DBStd       | 7.22   | 2.45  | 3.16  | 3.44  | 3.55  | 2.87  | 2.09  | 1.64  | 1.63  | 1.81  | 2.80  | 2.82  | 2.48  |
|                                                                   | HDD10.0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                   | HDD18.3     | 242    | 105   | 52    | 9     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 6     | 69    |
|                                                                   | CDD10.0     | 5724   | 158   | 206   | 352   | 491   | 642   | 689   | 733   | 728   | 642   | 542   | 340   | 201   |
|                                                                   | CDD18.3     | 2925   | 6     | 24    | 103   | 241   | 383   | 439   | 475   | 470   | 392   | 283   | 96    | 12    |
|                                                                   | CDH23.3     | 44227  | 111   | 372   | 1303  | 3255  | 5918  | 7203  | 7902  | 7681  | 5822  | 3498  | 992   | 169   |
| Wind                                                              | WSAvg       | 1.9    | 1.5   | 1.8   | 2.1   | 2.2   | 2.2   | 2.3   | 2.3   | 2.2   | 1.9   | 1.6   | 1.4   | 1.3   |
|                                                                   | PrecAvg     | 2      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0     |
|                                                                   | PrecMax     | 32     | 2     | 4     | 2     | 0     | 1     | 0     | 0     | 1     | 1     | 31    | 5     | 0     |
|                                                                   | PrecMin     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                   | PrecStd     | 6      | 0     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 6     | 1     | 0     |
| Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 28.8  | 33.1  | 37.8  | 42.2  | 44.8  | 45.2  | 44.5  | 44.2  | 42.7  | 40.2  | 34.6  | 29.9  |
|                                                                   |             | MCWB   | 18.4  | 18.7  | 20.8  | 21.5  | 22.9  | 23.6  | 24.1  | 24.6  | 23.7  | 23.0  | 21.2  | 19.0  |
|                                                                   | 2%          | DB     | 26.2  | 30.2  | 34.8  | 39.9  | 42.5  | 43.4  | 43.1  | 42.4  | 41.0  | 38.1  | 32.1  | 27.1  |
|                                                                   |             | MCWB   | 16.3  | 17.4  | 19.2  | 20.8  | 22.1  | 22.9  | 23.4  | 23.8  | 23.1  | 22.4  | 20.0  | 17.3  |
|                                                                   | 5%          | DB     | 24.1  | 27.8  | 32.1  | 37.6  | 40.9  | 42.1  | 41.9  | 41.2  | 39.8  | 36.4  | 30.3  | 25.0  |
|                                                                   |             | MCWB   | 15.1  | 16.1  | 17.9  | 19.9  | 21.3  | 22.5  | 23.0  | 23.3  | 22.9  | 21.8  | 19.1  | 16.1  |
|                                                                   | 10%         | DB     | 22.1  | 25.2  | 29.9  | 35.4  | 39.1  | 40.8  | 40.8  | 40.2  | 38.2  | 34.9  | 28.8  | 23.2  |
|                                                                   |             | MCWB   | 13.9  | 14.8  | 16.7  | 18.7  | 20.6  | 22.1  | 22.8  | 23.0  | 22.4  | 21.4  | 18.3  | 15.0  |
|                                                                   | 0.4%        | WB     | 19.9  | 19.8  | 22.0  | 22.7  | 23.7  | 24.7  | 25.5  | 26.0  | 25.5  | 24.4  | 22.5  | 20.5  |
|                                                                   |             | MCDB   | 26.0  | 30.8  | 34.1  | 39.3  | 41.9  | 42.4  | 41.8  | 41.0  | 39.9  | 37.0  | 31.9  | 28.0  |
| Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 17.0  | 18.1  | 19.9  | 21.4  | 22.7  | 23.8  | 24.5  | 24.8  | 24.2  | 23.3  | 20.8  | 17.9  |
|                                                                   |             | MCDB   | 25.2  | 29.3  | 32.6  | 37.5  | 40.7  | 41.4  | 40.7  | 40.4  | 38.6  | 36.0  | 30.6  | 25.7  |
|                                                                   | 5%          | WB     | 15.4  | 16.4  | 18.5  | 20.3  | 21.9  | 23.1  | 23.8  | 24.1  | 23.4  | 22.4  | 19.7  | 16.5  |
|                                                                   |             | MCDB   | 23.3  | 26.8  | 31.0  | 36.1  | 39.3  | 40.4  | 39.9  | 39.7  | 37.7  | 34.9  | 29.4  | 24.3  |
|                                                                   | 10%         | WB     | 14.2  | 15.0  | 17.1  | 19.2  | 21.0  | 22.4  | 23.2  | 23.5  | 22.8  | 21.6  | 18.6  | 15.3  |
|                                                                   |             | MCDB   | 21.8  | 24.7  | 29.3  | 34.6  | 37.9  | 39.2  | 39.0  | 38.9  | 37.0  | 33.7  | 27.8  | 22.7  |
| Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 14.7  | 15.3  | 15.7  | 16.3  | 15.8  | 15.4  | 14.6  | 14.5  | 14.9  | 14.6  | 14.7  | 14.6  |
|                                                                   |             | MCDBR  | 17.2  | 18.3  | 18.4  | 18.7  | 17.8  | 16.9  | 15.7  | 15.3  | 16.1  | 15.7  | 15.9  | 16.2  |
|                                                                   |             | MCWBR  | 10.1  | 9.8   | 9.1   | 8.5   | 7.7   | 7.2   | 6.6   | 6.4   | 6.9   | 7.3   | 8.4   | 9.3   |
|                                                                   | 5% WB       | MCDBR  | 16.7  | 17.7  | 17.5  | 18.0  | 16.7  | 16.0  | 14.6  | 14.7  | 15.1  | 15.0  | 15.4  | 15.8  |
|                                                                   |             | MCWBR  | 10.2  | 9.8   | 9.0   | 8.4   | 7.5   | 7.0   | 6.5   | 6.4   | 6.7   | 7.2   | 8.4   | 9.3   |
| Clear-Sky Solar Irradiance                                        | taub        |        | 0.375 | 0.408 | 0.451 | 0.537 | 0.551 | 0.473 | 0.484 | 0.488 | 0.469 | 0.466 | 0.392 | 0.372 |
|                                                                   | taud        |        | 2.164 | 2.045 | 1.918 | 1.733 | 1.721 | 1.927 | 1.929 | 1.926 | 1.952 | 1.946 | 2.138 | 2.175 |
|                                                                   | Ebn at Noon |        | 864   | 862   | 844   | 781   | 768   | 825   | 815   | 811   | 816   | 792   | 838   | 849   |
|                                                                   | Edh at Noon |        | 131   | 159   | 191   | 236   | 238   | 192   | 192   | 192   | 182   | 174   | 134   | 125   |
| All-Sky Solar Radiation                                           | RadAvg      |        | 4.54  | 5.47  | 6.58  | 7.39  | 7.82  | 8.17  | 7.91  | 7.46  | 6.77  | 5.70  | 4.78  | 4.25  |
|                                                                   | RadStd      |        | 0.16  | 0.19  | 0.20  | 0.14  | 0.20  | 0.09  | 0.10  | 0.11  | 0.15  | 0.12  | 0.09  | 0.08  |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                        | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only           | +0.67 | +0.60   | N/A    | +0.37   | +0.50 | +0.52 | N/A         | -57     | +256    | +204    |
| (47) | Regional (4 neighbors) | +0.65 | N/A     | N/A    | +0.49   | +0.42 | +0.41 | N/A         | -38     | +248    | +223    |

Nomenclature: See separate page